

THE FUTURE OF TRUST IS A TEAM SPORT

Authors:

Dominic Trott, Radoslav Dragov

May 2020

Sponsored by



IDC #EUR146278820



The Future of Trust Is a Team Sport

Introduction

Digital transformation (DX) is the fourth industrial revolution and like preceding revolutions requires fuel to succeed.

DX is increasingly reliant on the data generated by "connected things," yet the quality and provenance of this data varies widely. To optimize DX, and to support board-level business decisions, the data from connected things must be truthful. But it is difficult to establish trust with wildly varying approaches to vulnerability, security, and configuration management. If data is the "new oil," it needs some refinement.

Distributed ledger technology (DLT) is proving its value in partner ecosystems by providing all stakeholders with end-to-end visibility, while guaranteeing data integrity. It is critical to note that these value chains are not limited to supply chains for physical goods, but rather span the collaboration between stakeholders on a much broader and ongoing basis. In this way, building trust in connected "things" is a team sport with each member of the connected device ecosystem playing a role to drive mutual benefit.

AT A GLANCE

KEY STATS

- » Digital trust is the top new agenda priority for CxOs for the next five years.
- » 79% of enterprises do not audit the security risk of all their supply chain partners.
- » Quantifying the business benefits of trust is the top digital trust priority for CxOs.
- » Over 70% of manufacturers rate digital twins as an important strategic initiative.
- » By 2023, IDC predicts that 65% of global manufacturers will have realized operational expense savings of 10% through process digital twins.
- » The percentage of enterprises that have adopted/plan to adopt blockchain rose from 18% in 2018 to 34% in 2019.
- » IDC forecasts there will be 55.8 billion "connected things" by 2025, up from 40.4 billion in 2020.

Digital Trust Is the Number 1 New Agenda Priority for CxOs

IDC's research shows digital trust is the top priority for CxOs for the next five years, and for good reason (source: IDC, *The CEO's Agenda for the Digital Enterprise*, 2019). DX is ubiquitous and is the heart of business strategy for 89% of enterprises (source: IDC, *Understand How Organizations Evolve to a Digital Business Model*, 2018). Yet DX requires conscious management and innovative ways of collaborating both internally and externally if its benefits are to be realized.

A key example is the convergence of technology. Here, we increasingly see information technology (IT), operational technology (OT), the Internet of Things (IoT), and industrial IoT (IIoT) being drawn together into large hybrid systems that are expected to simply work together out of the box. However, these technologies grew up separately and have inherently different and separate modes of operation, design, and security. This issue is exacerbated when considering the explosion of the number of connected devices, transforming the nature and scale of transactions with partners and customers in addition to colleagues.

Additionally, inconsistent levels of testing, patch management, and vulnerability management across the various technology components of today's smart infrastructure environments make it extremely difficult to assure trust in any kind of consistent (let alone programmatic) fashion, often reliant on manual processes and intervention.

DX cannot be achieved, then, without significant collaboration between the IT team and operational units, and processes that satisfy both sides that things are "being done right."

79% of Enterprises Do Not Audit Security Risk for All Their Partners; 32% Take No Steps

DX drives a huge expansion of data generation, storage, and exchange. IDC's research shows that the volumes of data generated by connected things every year are measured by the zettabyte (i.e., 1,000 to the power of 7 bytes). Yet there is a "missing ingredient" for these enterprise transactions: 79% of enterprises do not audit the security risk of their partner ecosystem and 32% take no steps whatsoever to audit partner security risk (source: IDC, *European Security Strategies Survey*, 2019).

Even when partner risk management and trust development programs have been put in place, they tend to be inconsistent, fragmented, and dependent on manual processes. That goes for internal approaches for a single enterprise, therefore the problem is magnified when aiming to distribute trust across an ecosystem of partners.

In IDC's view, rather than an inconsistent mixture of approaches that vary between supply chain members, there is an opportunity to drive a consistently strong level of confidence by adopting a distributed, automated approach to digital trust.

This way, partners are encouraged to collaborate for mutual benefit, driving up standards across the board. For example, among partners that work in critical infrastructure environments, the adoption of a collaborative cybersecurity maintenance platform can help to turn third-party involvement from a perceived source of risk into a demonstrable strength.

Quantifying Business Benefits Is the Number 1 Trust Priority for CxOs

Paradoxically, with trust sitting atop the board's agenda, IDC's research shows that the top trust challenge for enterprises is measuring the business benefits of trust (source: IDC, *CxO View of the Future Enterprise in the Digital Economy*, 2020). From one perspective this is understandable, as trust is an abstract and nebulous concept. But in IDC's view, the benefits are manifold: trust accelerates speed of operations, reduces cost, and boosts efficiency.

While perception remains a critical external influence, it is trust that greases the gears of the digital economy. With so much data being generated and transacted, compounded by the acceleration of connected things being deployed, this simply cannot happen at the speed and scale required to maximize the opportunity of DX without a foundation of automated and reliable trust mechanisms.

TABLE 1
The Business Benefits of Digital Trust

Security risk visibility	Drive visibility into the history of connected machines and the data they generate to enable infrastructure stakeholders to understand the risks and the resulting impacts that connected things may represent for the enterprise.
Reduce risk through collaboration	Respond to risks. A distributed team can be built, spanning the various stakeholders, that will be able to move fast based on that visibility to fix issues.
Report to regulated industries	Collaborative maintenance platform can continually report compliance on the security and safety of connected things.
Reduce costs through automation	The ability to automate outcomes through smart contracts — as enabled by distributed ledgers — reduces the requirement for manual intervention. This releases time-poor security personnel to focus on higher-value tasks rather than mundane compliance reporting.
Reinforce and enrich data	Data integrity and provenance via DLT lets collaborative maintenance platforms guarantee the truthfulness of machine data. This means business leaders can make faster, better decisions as a result of assured intelligence.
Knock down the barriers to DX	Demonstrating the provenance of connected machine data is valuable internally to direct stakeholders. But it is also valuable to third parties that may wish to use the data for data-driven business models, such as cyberinsurance providers.

Source: IDC, 2020

IDC believes that the "secret sauce" for demonstrating the value of trust will be to focus on the automation of tasks that currently tend to be conducted manually but which are mission critical in nature. The promise of automation here is to save time for security teams, enabling them to do more for less. There are six primary avenues through which a collaborative maintenance platform can drive these benefits, as shown in Table 1. The key to unlocking mounting interest in digital trust to become a driver for change will be to demonstrate precisely where those business benefits lie.

Benefits

The Future of Trust

Going Beyond "Zero Trust"

When it comes to the language of trust, at present the market tends to coalesce around the concept of zero trust. This is intended to describe a more powerful and granular model for access control than typical perimeter-based approaches, involving rules for users accessing applications that better model risk.

Zero trust is a useful concept, and the adoption of a "default deny" approach, beyond which specific access is allowed only as it is granted, helps build integrity across the enterprise. However, the interconnected nature of smart infrastructures and the digital economy means

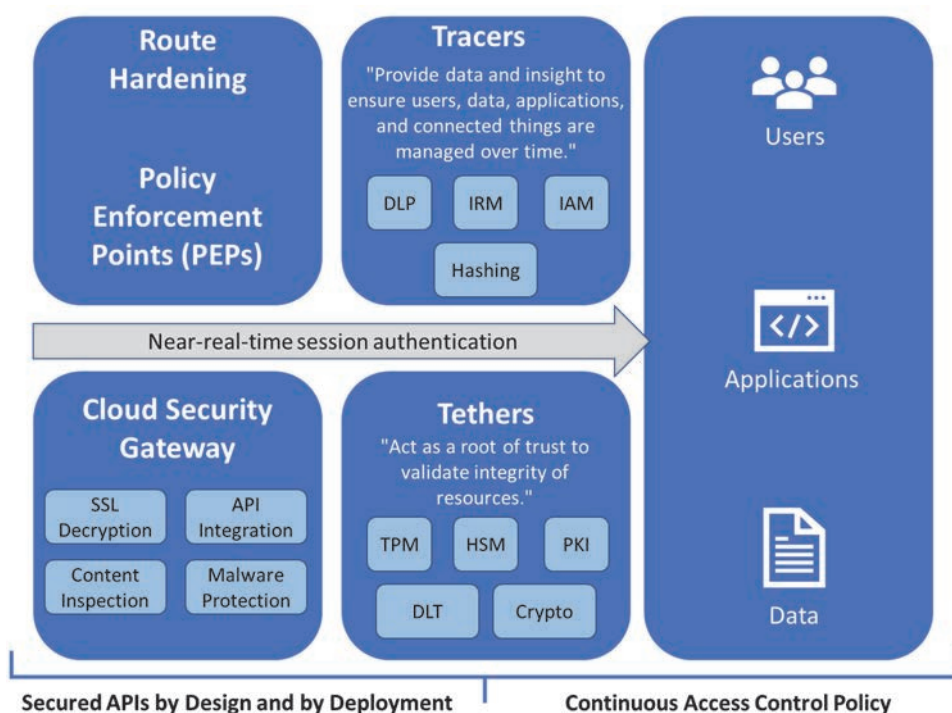
that integrity must also be distributed across the partner ecosystem. Therefore IDC extends and underpins zero trust with a concept it terms distributed integrity.

IDC defines distributed integrity as "a cybersecurity architectural model that is focused on how users and devices access applications and process data. Within this model are two common capabilities — software-defined perimeters (SDPs) and microsegmentation — that comprise most technical approaches to zero trust" (source: IDC, *Microsegmentation for "Zero Trust" Distributed Integrity*, 2019).

Distributed Integrity to Build Collaborative Trust

FIGURE 1

Distributed Integrity Model



Source: IDC, 2020

Given its superior expressive power and focus on business outcomes, distributed integrity is a vital concept when considering how to extend trust and integrity beyond the enterprise as an individual entity into the partner ecosystem.

Figure 1 provides a conceptual map with sample technologies to demonstrate how a distributed integrity approach can work. Critically for collaborative trust, the distributed integrity model provides for two key concepts: "tracers" provide insight into how data, applications, and connected things are managed over time, and "tethers" that offer a "root of trust to validate the integrity of resources." Together, these provide a foundation that can offer a common view of what underpins trust between enterprise partners.

Traditionally, larger and more mature enterprises have adopted the use of technologies such as HSMs and PKI to achieve a level of trust in the integrity of assets and identities that they control

and depend on. Such choices represent an ideal for siloed operations or industries where a small number of powerful players dictate the rules for a much larger group (such as card payments or big finance). But they do not provide the answer for distributed trust where enterprises need to trust *each other* at a deeper level and establish a level playing field of trust, security, and accountability for their joint activities.

Beyond the simple fact that the costs and labor required are prohibitive and therefore exclusive to many smaller organizations, inconsistent technology choices and management processes are still an uncontrollable risk to the group as a whole. Further, when a vulnerability does arise, the large amount of faith placed in one basket can be worse than having no protection at all.

But crucially, with these traditional solutions each enterprise holds the master controls necessary to cover its tracks, or falsify records, or deny the word of the group. They lack the *ambient accountability* that is vital to implementing true distributed integrity.

Therefore, with any theoretical "ideal" out of reach for vaulting or secrecy-based solutions, there needs to be a level of hygiene in place when it comes to building trust in connected devices: tracers. To this end, DLT-based solutions represent a logical first step, tracing visibility into device history and data that enable an understanding of device trustworthiness and machine data provenance. This in turn will enable businesses to make decisions based on machine data with confidence. Once that is achieved, then they can make targeted choices on where to invest in stronger technology and where to invest in process.

Extending the Concept of Digital Twins to Security Twins

What Is a Digital Twin?

Over 70% of manufacturers rate digital twins as an important strategic initiative. IDC defines a digital twin as "a virtual model of a product, identity, or asset connected to the related physical prototype or instance via IoT, thus visualizing data flow, communication, and collaboration across engineering, operations, the supply chain, and servicing" (source: IDC, Opportunities for Digital Twins in Manufacturing Business Ecosystems, 2020). A collaborative maintenance platform is a good example of where this concept can be applied.

As well as being a "strategic" technology, digital twins offer significant business benefits. For example, IDC predicts that by 2023 65% of global manufacturers will have realized savings of 10% in operational expenses through process digital twins driven by IoT and machine learning routines that factor in unstructured data sets (source: IDC, FutureScape: Worldwide Manufacturing 2020 Predictions, 2019).

What Is a Security Twin?

The idea of a security twin is foundational to the trustworthiness of the data generated by connected devices, and especially where reported by their digital twins. A security twin fulfills the role of a "tracer" in the distributed integrity model. It is based on three elements:

1. A consistent, communally approved view of the identity and shared service history of an asset. This provides a record of who did what to a thing and when.

2. A shared service history that records attributes of connected devices such as updates, vulnerabilities, configurations, and compliance — giving an indication of its trustworthiness.
3. A system that can automate workflow among participants of a smart infrastructure — and the connected devices of which it comprises — to move fast and fix things.

Multiple benefits are provided through this approach, but this document has identified three key themes. First, a security twin provides a foundation that all stakeholders of a smart infrastructure can trust and understand. It gives owners, maintainers, auditors, manufacturers, and systems integrators a platform that enables them to quickly collaborate with confidence to drive mutual benefit.

Second, security twins can automate the "tagging" of data that originates from untrusted and/or non-compliant sources. In this way, enterprises are empowered to handle tagged data depending on circumstances and context. For example, rather than simply deleting untrusted data outright, it may be that a secondary source can be checked to verify trust in the data itself or trigger a reauthentication request for machine sending data. Significantly, this benefit can be delivered before data spreads more widely within the enterprise. In both cases, the automation of this activity offers significant efficiencies.

Third, security twins provide a mechanism for demonstrating trust in the data generated by digital twins. Without the need for costly individual integrations, any enterprise, application, or person relying on digital twin data can operate safe in the knowledge that the data is provable as being secure, compliant, and trustworthy based on known sources. In much the same way that identity and access management (IAM) technologies help to protect enterprise data by authenticating users and validating document signatures before they come together, so security twins take care of verifying IoT data sources and digital twin access before any critical business process acts on that data.

Through these attributes, security twins offer the promise of helping enterprises to pursue their DX strategies securely and based on a foundation of trust. This move can be achieved incrementally if required. In fact, taking a "rip and replace" approach to existing infrastructure to meet trust requirements is likely to be both unaffordable and unpopular. Significantly, though, taking a gradual approach to infrastructure replacement can help move enterprises toward the position of being able to tell the difference between older, unproven data and newer, more trustworthy data whose provenance is clear. This in turn can support the incremental improvement of operations linked to connected things.

It is important to bear in mind the multiparty nature of DX and the interconnected nature of digital supply chains. To meet this multiparty reality, a multiparty solution is required. This is where the power of security twins is made particularly evident, facilitating collaboration via distributed integrity models.

Automation — The Catalyst for Digital Trust Programs

Problems with Current Practice

IDC's research indicates that 97% of enterprises not only agree that there is a security skills shortage, but that this has negative consequences (source: IDC, European Security Strategies Survey, 2018). Using typical approaches, the operation of trustworthy smart infrastructures and connected devices is highly labor intensive. Research shows that the typical manufacturer operates over 500 different IoT devices from over 200 different manufacturers (source: Zscaler, IoT in the Enterprise, 2020). Given that the skills for converged IT/OT security are even more scarce, the scale of the problem becomes clear.

To effectively manage and maintain such an environment while fully accounting for and mitigating the multiple vulnerabilities, configuration needs, and compliance requirements would be a feat worthy of Sisyphus — the mythical Greek king punished to an eternity of rolling a boulder up a hill only to have it roll back down again. Security is never "complete" — it requires constant vigilance, and to create such a Sisyphean team would call for a significant investment, with few organizations having the necessary resources.

IDC's research shows that the average enterprise has a security team made up of 5.7 full-time employees (source: IDC, Security Policy Survey, 2019). It is reasonable to estimate that, for this average security team, the job of continually updating the understanding of expanding enterprise infrastructure, testing those devices for vulnerabilities, prioritizing and responding accordingly, and on top of that identifying and deploying all known patches, would require that team of approximately 6 to be doubled to 12.

With security teams already struggling for resources, doubling the team would be a massive leap in security budget. Therefore, in reality this task is: 1) never fully delivered and 2) dropped on the security team's desk rather than funding a separate infrastructure trust team. The net result of both is to escalate the growing problem of burnout in the security team, a further limitation of operational efficiency and resource utilization.

Opportunities from Security Twins

Current practice is neither enabling trust nor maximizing operational effectiveness. Security twins present an opportunity to deliver on both goals through the implementation of technology that can automate manual processes, reduce pressure on security professionals, and enhance trust. Security twins can help security teams to automate tasks in the following three key areas:

- **Asset monitoring and reporting** — providing a close-to-real-time view of the trust status of connected devices
- **Risk mitigation** — delivering risk-based prioritization of security and maintenance workflow to mitigate threats through vulnerability management
- **Smart contracts** — a means of automating SLA-based responses, sequence actions, scheduling remedies, and raising alerts that is provable and acceptable to all stakeholders

This ability to reduce pressure on the workforce while simultaneously improving trust outcomes is critical when considering the existing resource pressure placed on security teams. Therefore, the opportunity to deliver more for less through security twins is timely.

What is more, there are broader opportunities that emerge from the security twin approach. As identified above, there are six key business benefits that can be targeted through digital trust as enabled by a collaborative maintenance platform. To give a specific example, third parties are likely to show interest in machine data that benefits from proof of origin and integrity to unlock new business models.

For example, in much the same way that a car telemetry box enables car insurers to reduce premiums for safe drivers, cyberinsurance providers can offer reduced premiums to stakeholders in smart infrastructure deployments that can, for example, demonstrate regulatory compliance and provide a rigorous approach to patch management and vulnerability remediation.

Trends

The Power of Distributed Ledger Technologies

Distributed ledger technology (DLT) is an umbrella term that is used to describe technologies that record data in multiple places at the same time. In the development of a security twin solution, organizations should embrace the characteristics of a private permissioned ledger: an incorruptible, decentralized network with enhanced security properties.

A few years ago, DLT emerged with high expectations about the transformative potential of the technology. In the IDC European Tech and Pulse Industry Survey (which samples nearly 2,800 respondents) the percentage of respondents who have adopted or plan to adopt blockchain increased from 18% to 34% in just one year, between 2018 and 2019. While DLTs can be applied in many use cases across industries, there are specific applications where the technology is most relevant and brings the most tangible benefits. As a decentralized "trustless" network, DLT is best applied to use cases with many stakeholders who have diverging interests and seek ways to distribute trust. One of the most popular DLT applications has been in supply chains due to the ability to create a secure collaborative system between multiple parties without a trusted third-party intermediary. The same benefits apply to cybersecurity partner ecosystems, which are often very vulnerable — especially when there are many stakeholders exchanging sensitive information.

No Single Point of Failure

Unlike in traditional databases, in a distributed ledger (DL) the information isn't held in one place and all authorized members have access to the ledger. The system is connected and synchronized among all members of the network who reach a consensus on a single version of history. DLs remove the risks associated with information being stored in one place and validated by one actor, namely that information can be easily hacked or that the party who controls the central database may retroactively change information. The decentralized nature of a DL ensures there is no single point of failure, guaranteeing stability and high uptime.

Immutability

DLs create a permanent record of transactions that are cryptographically sealed. No member of the network can change information once it's recorded on the distributed ledger. Users who post incorrect data have the ability to post subsequent information, but the original record will remain and cannot be expunged. In case of a dispute between members of the network the two sides can review the audit trail with the knowledge that no party can game the system and change already recorded information.

Breaking Down Data Silos

A widespread problem with data sharing is that companies are completely siloed from each other and use different (legacy) solutions to store data. There's a lot of effort and expense in transferring and translating information from one system to another. The problem is compounded when you have a long chain of companies exchanging data. By using DLs, all stakeholders are part of the same ledger and can share data in real time. There's no need for a third party to arbitrate the version of events. Exchange of information is more efficient, less costly, and less prone to errors.

Accountability and Trust

The use of DLs has a direct influence on the behavior of participants in the network. It fosters trust because no party has total control of the system and no one can cheat without leaving a permanent mark. The atmosphere of trust and cost-effective data exchange incentivizes even greater transparency and data sharing. DLs can be an indispensable component of any solution that tries to ensure fast, efficient secure data exchange.

Regulatory Reform

Regulatory reform is usually an accelerator for technology deployments to support the achievement and demonstration of compliance. This applies equally to security, where compliance is among the top priorities. IDC's research shows that compliance is the fourth-strongest priority for security decision makers, behind only security culture and awareness, managing user identities and access, and IoT security (source: IDC, European Security Strategies Survey, 2019).

There are many regulations emerging across multiple geographies that have an impact. Examples include California's SB-327 law covering information privacy across connected devices and the U.S. federal government's 2018 Cybersecurity and Infrastructure Security Agency Act, which (among other measures) established the Cybersecurity and Infrastructure Security Agency (CISA) itself, which is empowered to conduct "specialized field assessments to identify vulnerabilities, interdependencies, capabilities, and cascading effects of impacts on the nation's critical infrastructure." However, the furthest reaching (in terms of both legislative impact and its supra-national reach) example of IoT security law is the EU's Network and Information Systems Directive (NISD). The NISD accelerates the urgency and beneficial impact that security twins represent. To offer context, from a very high-level perspective, the NISD has impact in the following three areas:

- It is targeted at the operators of critical infrastructure in the EU, the definition of which has two groups. First there are the typical "operators of essential services" (OESs) such as energy, transport, and health providers. On top of this, the NISD extends the term of critical infrastructure to apply to what it terms "digital service providers" (DSPs) such as search engines, online marketplaces, and cloud computing services.
- Both OESs and DSPs are required to secure their network and information systems by taking technical and organizational measures appropriate to the risk they face.
- OESs and DSPs are also required to ensure service continuity by taking appropriate measures to prevent and minimize the impact of any incident.

Security twin concepts will be critical for both OESs and DSPs to facilitate trust and collaboration across the connected devices that make up smart infrastructure. Not only in the sense of improving the speed with which enterprise partners can make decisions based on connected devices, but also in terms of reducing the risk that these devices represent.

Importantly, regulatory reform such as the NISD shows that an ongoing focus on outcomes — rather than discrete, "tick-box" exercises — represent the future of compliance in this field. Hiding behind willful ignorance, plausible deniability, or prioritizing shareholder value over accumulated risk will not wash. Not least as regulations such as the NISD are armed with "teeth," being enforceable with fines. Therefore, the goal must be continual compliance and transparency.

Challenges

DX — The Disruptive Influence of Our Time

DX is not just an incremental shift in the way that enterprises work and operate, such as switching from paperwork to PDFs while retaining legacy processes. Rather, DX is a true transformation through data-driven applications. The benefits are huge, as demonstrated by IDC's research indicating that DX is central to the business strategies of 89% of enterprises. The shift toward a data-driven economy supports improved decision making, drives operational efficiency, and opens new markets, just to give a few examples.

While the rise of DX presents many opportunities, it also represents a step change from a risk perspective. The numbers are phenomenal. IDC expects that 50 zettabytes (ZBs) will be generated in 2020, expanding to 79.4ZB by 2023 (source: IDC, [Worldwide Global DataSphere IoT Device and Data Forecast, 2019–2023](#), 2019). This represents a massive extension to the attack surface, with every piece of data used by enterprises representing a potential target for threat actors.

Connected Devices Increase Exposure to Risk

While the growing volumes of data driven by DX strategies represent an expansion of the attack surface, this is extended through the proliferation of connected devices. IDC forecasts that there will be 55.8 billion "connected things" by 2025, up from 40.4 billion in 2020. That spans all of IT, OT, IoT, and IIoT (source: IDC, [Worldwide Global DataSphere IoT Device and Data Forecast, 2019–2023](#), 2019).

The vulnerabilities of these devices, combined with varying enterprise approaches to patch management, vulnerability management, and maintenance, adds to the risk. Inevitably, the multiparty nature of connected device deployments extends these risks beyond the enterprise perimeter and across the partner ecosystem.

The Double-Edged Sword of Digital Twins

While representing another pillar of the data-driven economy, digital twins themselves create an exposure to risk. This primarily relates to the risk of making decisions based on incomplete, incorrect, or even falsified data. What if the devices that generate digital twin data have been incorrectly deployed or configured? What if a malicious third party is willfully tampering with those devices and/or the data generated by them?

These are all new risks added to the enterprise through digital twin use. Again, the interconnectedness of smart infrastructures and DX means that any single enterprise's exposure to digital twin risk is repeated across the supply chain.

Conclusion

The rise of DX driven by data from connected devices increases the interconnectedness and interdependence of enterprises across their technology supply chains. This raises the risk of making damaging business decisions on data from devices of unknown provenance.

Manually developing a truly distributed trust model cannot scale, but as DX and connected device deployments expand, the criticality of trust will mount, requiring action from already stretched security teams.

A logical solution to this problem is the deployment of security twins, facilitating distributed integrity and a trust model that enables collaboration across DX and smart infrastructures while also reducing the burden on scarce, over-burdened security teams. The business case behind this technology is evident. Clearly, security twin concepts are indicative of the future of trust.

Vendor Profile — Jitsuin

Jitsuin was founded in 2018 by three industry veterans with decades of experience between them in deeply embedded silicon chip security, trusted execution environments, cryptography, software development, digital identity, and distributed ledger technology. Despite the widespread availability of security and trust technologies within connected things, the founding team saw many were overlooked or underutilized, and all data ended up being treated the same — dumped in a silo with the lowest common denominator of trust. Digital transformation with connected things flies blind with no visibility of data provenance.

The rise of connected things in critical infrastructures and operational technologies compounds business risk from vulnerabilities and exposures with the possibility of physical harm and real-world disruption that raise the stakes in cybersecurity and business continuity for operators of essential services.

Cybersecurity of connected things is a continuous delivery supply chain. All members of the supply chain have their own responsibilities and risks to manage, but they can't operate alone. Makers of connected things patch vulnerabilities. System integrators test for compatibility. Owners approve patches or otherwise mitigate. Maintenance teams schedule downtime to safely apply patches. Auditors keep account of service level agreements and regulators oversee organizations to build confidence in the system. The cybersecurity supply chain needed a new, transparent, collaborative, and automated platform to underpin trust in digital transformation with connected things. Jitsuin saw the opportunity to apply distributed ledger technology to provide the foundation for this supply chain collaboration and distributed integrity.

Jitsuin Archivist is aimed at all enterprises digitally transforming with connected things, but particularly in manufacturing, energy and utilities, transportation, buildings and infrastructure, healthcare, finance and insurance, platforms, and digital services. It enables CISOs to do more with less:

- Enable trust in DX — for the business to build new revenue and seek greater value
- Reduce risk from connected things — by enabling a multiparty collaborative team to automatically reveal and resolve issues while continually reporting regulatory compliance

Jitsuin Archivist is a SaaS platform built with DLT that stakeholders use to permanently record who did what to a thing and when. Those transactions form a high integrity shared service history for each connected thing — its security twin. Information about how connected devices protect their identities, what they are built from, and how they have been configured and maintained throughout their lifetime all add up to a proven track record for data from each connected thing.

Jitsuin Archivist security twins:

- Reduce administrative overheads
- Enrich data with provenance, reinforce integrity, and enable faster, better business decisions

Things are only secure until they are not; that's one certainty for a predictive maintenance doctrine. Prescriptive cybersecurity maintenance routines encoded in automated smart contracts enable you to move fast and fix things. With an automated ledger of cybersecurity maintenance, compliance audits evolve from one-off adversarial exercises to continual cryptographic proof of security and safety in regulated industries.

Jitsuin Archivist is available on [Azure marketplace](#) as a SaaS subscription. It is a data-first platform that integrates with REST APIs to any other data sources — IoT platforms, IAM systems, and business applications. It requires no specialist knowledge of DLTs for enterprise integration and no device-level agents to be installed. Jitsuin means "truth in things."

About the Analysts



[Dominic Trott](#), Research Director, European Security and Privacy

Dominic Trott is the research director for IDC's European Security and Privacy domain. As well as managing this team of analysts, Trott runs IDC's European CISO outreach program, chairing IDC's European CISO Advisory Board and the European CISO Hub panel. Trott focuses on the top challenges for European security practitioners, including the evolving role of the CISO, security business outcomes, operational excellence, and the future of trust.



[Radoslav Dragov](#), Senior Research Analyst, European Customer Insights and Analysis, European Blockchain

Radoslav Dragov is a senior research analyst for IDC's European Customer Insights and Analysis team. Based in London, he leads the IDC European Blockchain Practice and research. Through both ongoing syndicated and custom research, he helps IT vendors and end users understand the impact of blockchain and distributed ledger technology (DLT) on their business plans and digital strategy.

About IDC

International Data Corporation (IDC) is the premier global provider of market intelligence, advisory services, and events for the information technology, telecommunications, and consumer technology markets. IDC helps IT professionals, business executives, and the investment community make fact-based decisions on technology purchases and business strategy. More than 1,100 IDC analysts provide global, regional, and local expertise on technology and industry opportunities and trends in over 110 countries worldwide. For 50 years, IDC has provided strategic insights to help our clients achieve their key business objectives. IDC is a subsidiary of IDG, the world's leading technology media, research, and events company.

IDC UK

5th Floor, Ealing Cross,
85 Uxbridge Road
London
W5 5TH, United Kingdom
44.208.987.7100
Twitter: @IDC
idc-community.com
www.idc.com

Global Headquarters

5 Speen Street Framingham, MA
01701 USA
P.508.872.8200
F.508.935.4015
www.idc.com

Copyright and Restrictions

Any IDC information or reference to IDC that is to be used in advertising, press releases, or promotional materials requires prior written approval from IDC. For permission requests contact the Custom Solutions information line at 508-988-7610 or permissions@idc.com. Translation and/or localization of this document require an additional license from IDC. For more information on IDC visit www.idc.com. For more information on IDC Custom Solutions, visit http://www.idc.com/prodserv/custom_solutions/index.jsp.

Copyright 2020 IDC. Reproduction is forbidden unless authorized. All rights reserved.